

MSTAR PRACTICE WORKSHEET**Lesson 5-8: Determine Surface Area of Pyramids**

Grade 6 Mathematics · Standard 6.GR.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

- ☐ A 70 in²
- ☒ B 95 in²
- ☐ C 95 in³
- ☐ D 120 in²

Base: $5 \times 5 = 25$ in². Each lateral face: $\frac{1}{2} \times 5 \times 7 = 17.5$ in². Four faces: $4 \times 17.5 = 70$ in². SA = $25 + 70 = 95$ in².

Question 2 **SELECTED RESPONSE — CORRECT: B**

How many lateral (triangular) faces does a square pyramid have?

- ☐ A 3
- ☒ B 4
- ☐ C 5
- ☐ D 6

A square pyramid has a square base and 4 triangular lateral faces — one for each side of the square.

Question 3 **SELECTED RESPONSE — CORRECT: B**

What is the area of one triangular face with base 8 in and slant height 6 in?

- ☐ A 14 in²
- ☒ B 24 in²
- ☐ C 24 in³
- ☐ D 48 in²

Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 6 = 24 \text{ in}^2$.

Question 4 **SELECTED RESPONSE — CORRECT: C**

A square pyramid has a base area of 49 cm² and a total lateral area of 84 cm². What is the surface area?

- ☐ A 49 cm²
- ☐ B 84 cm²
- ☒ C 133 cm²
- ☐ D 133 cm³

SA = Base Area + Lateral Area = 49 + 84 = 133 cm².

Question 5 **SELECTED RESPONSE — CORRECT: C**

Your first gift box is a square pyramid with a base edge of 7 in and a slant height of 9 in. How much wrapping paper (total surface area) covers it?

- ☐ A 126 in²
- ☐ B 144 in²
- ☒ C 175 in²
- ☐ D 175 in³

Base: $7 \times 7 = 49 \text{ in}^2$. One triangular face: $\frac{1}{2} \times 7 \times 9 = 31.5 \text{ in}^2$. Four faces: $4 \times 31.5 = 126 \text{ in}^2$. SA = $49 + 126 = 175 \text{ in}^2$. Surface area uses square units (in²).

Question 6

SELECTED RESPONSE — CORRECT: B

Before adding the base, you need the area of just one triangular face of a gift box with base edge 8 in and slant height 5 in. What is it?

- ☐ A 13 in²
- ☒ B 20 in²
- ☐ C 20 in³
- ☐ D 40 in²

Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 5 = 20 \text{ in}^2$. You use the slant height (5) as the triangle's height, not the pyramid's vertical height.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: C**

A square pyramid has a base measuring 6 m by 6 m and four triangular faces, each with a base of 6 m and a slant height of 5 m. What is its surface area?

- ☐ A 36 square meters
- ☐ B 60 square meters
- ☒ C 96 square meters
- ☐ D 156 square meters

Base area = $6 \times 6 = 36$. Each triangle = $\frac{1}{2} \times 6 \times 5 = 15$, and $4 \times 15 = 60$. Total = $36 + 60 = 96$ square meters.

- **A:** That is the square base alone. Add the four triangular faces.
- **B:** That is the four triangles alone. Add the 6 by 6 base.
- **D:** The triangles were never halved: four times 6 times 5 is 120 rather than 60. A triangle is half of base times height.

Part B — correct answer: A

Why does a pyramid's surface area use the slant height rather than the vertical height?

- ☒ A The slant height is the height of each triangular face, measured along the face itself.
- ☐ B The vertical height is always longer than the slant height.
- ☐ C The slant height is easier to measure.
- ☐ D Pyramids do not have a vertical height.

Each lateral face is a triangle lying on the surface, and its height is the slant height. The vertical height runs through the inside of the solid and is not a face measurement.

Question 8**SELECT ALL THAT APPLY — CORRECT: A, B, C, E**

Select ALL statements that are true about the surface area of a pyramid:

- ☒ A Surface area = base area + the areas of all lateral faces.
- ☒ B A square pyramid has 4 triangular lateral faces.
- ☒ C Each lateral face area is $\frac{1}{2} \times$ base of the triangle $\times$ slant height.
- ☐ D The vertical height is used to find each triangular face area.
- ☒ E A pyramid's net shows the base and every lateral face laid flat.

A, B, C, and E are true. D is false — the slant height, not the vertical height, is the height of each triangular face.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Two pyramids have the same base area (64 in^2) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

Model answer: With a 64 in^2 base the edge is 8 in, so Pyramid A has $SA = 64 + 4(\frac{1}{2} \times 8 \times 6) = 64 + 96 = 160 \text{ in}^2$ and Pyramid B has $SA = 64 + 4(\frac{1}{2} \times 8 \times 10) = 64 + 160 = 224 \text{ in}^2$, a difference of 64 in^2 . Slant height only appears in the triangle formula, so it stretches the lateral faces while the base area stays 8×8 no matter how tall the pyramid gets.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.